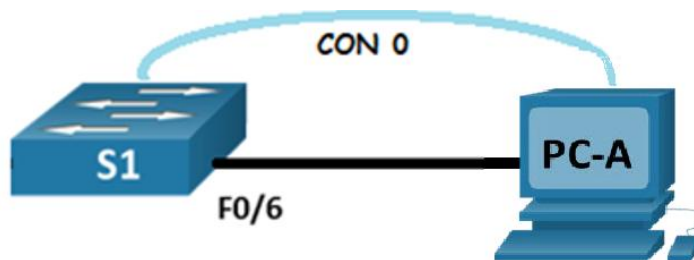


Lab 7.2.7 - View Network Device MAC Addresses

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
S1	VLAN 1	192.168.1.2	255.255.255.0	N/A
PC A	NIC	192.168.1.3	255.255.255.0	192.168.1.1

Objectives

Part 1: Configure Devices and Verify Connectivity

Part 2: Display, Describe, and Analyze Ethernet MAC Addresses

Background / Scenario

Every device on an Ethernet LAN is identified by a Layer 2 MAC address. This address is assigned by the manufacturer and stored in the firmware of the NIC. This lab will explore and analyze the components that make up a MAC address, and how you can find this information on a switch and a PC.

You will cable the equipment as shown in the topology. You will configure the switch and PC to match the addressing table. You will verify your configurations by testing for network connectivity.

After the devices have been configured and network connectivity has been verified, you will use various commands to retrieve information from the devices to answer questions about your network equipment.

Note: The switches used are Cisco Catalyst 2960s with Cisco IOS Release 15.2(2) (lanbasek9 image). Other switches and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Make sure that the switches have been erased and have no startup configurations. If you are unsure, ask your instructor.

Required Resources

- 1 Switch (Cisco 2960 with Cisco IOS Release 15.2(2) lanbasek9 image or comparable)
- 1 Laptop (with a terminal emulation program, such as Tera Term or Putty)
- 1 Laptop (Windows, Mac, or Ubuntu Linux)
- Console cable to configure the Cisco switch via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Configure Devices and Verify Connectivity

In this part, you will set up the network topology and configure basic settings, such as the interface IP addresses and device name. For device name and address information, refer to the Topology and Addressing Table.

Step 1: Cable the network as shown in the topology.

- a. Attach the devices shown in the topology and cable as necessary.
- b. Power on all the devices in the topology.

Step 2: Configure the IPv4 address for the Laptop (PC-A)

- a. Configure the IPv4 address, subnet mask, and default gateway address for PC-A.
- b. From the command prompt (or terminal) on PC, ping the switch address.

Were the pings successful? Explain.

The switch has no ip address so the ping as unsuccessful.

Step 3: Configure basic settings for the switch.

In this step, you will configure the device name and the IP address, and disable DNS lookup on the switch.

- a. Console into the switch and enter global configuration mode.

```
Switch> enable
Switch# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
```

- b. Assign a hostname to the switch based on the Addressing Table.

```
Switch(config)# hostname S1
```

- c. Disable DNS lookup.

```
S1(config)# no ip domain-lookup
```

- d. Configure and enable the SVI interface for VLAN 1.

```
S1(config)# interface vlan 1
S1(config-if)# ip address 192.168.1.2 255.255.255.0
S1(config-if)# no shutdown
S1(config-if)# end
*Mar  1 00:07:59.048: %SYS-5-CONFIG_I: Configured from console by console
```

Step 4: Verify network connectivity.

Ping the switch from the PC.

Were the pings successful?

The pings were successful.

Every device on an Ethernet LAN has a MAC address that is assigned by the manufacturer and stored in the firmware of the NIC. Ethernet MAC addresses are 48-bits long. They are displayed using six sets of

hexadecimal digits that are usually separated by dashes, colons, or periods. The following example shows the same MAC address using the three different notation methods:

00-05-9A-3C-78-00

00:05:9A:3C:78:00

0005.9A3C.7800

Note: MAC addresses are also called physical addresses, hardware addresses, or Ethernet hardware addresses.

You will issue commands to display the MAC addresses on a PC and a switch, and analyze the properties of each one.

Step 5: Analyze the MAC address for the PC NIC.

Before you analyze the MAC address on the PC, look at an example from a different PC NIC. You can issue the **ipconfig /all** command to view the MAC address of your NIC (note: Use the ifconfig command from the terminal on MAC and Ubuntu Linux PC). An example screen output is shown below. When using the **ipconfig /all** command, notice that MAC addresses are referred to as physical addresses. Reading the MAC address from left to right, the first six hex digits refer to the vendor (manufacturer) of this device. These first six hex digits (3 bytes) are also known as the organizationally unique identifier (OUI). This 3-byte code is assigned to the vendor by the IEEE organization.

To find the manufacturer, use the keywords **IEEE OUI standards** to find an OUI lookup tool on the internet or navigate to <http://standards-oui.ieee.org/oui.txt> to find the registered OUI vendor codes. The last six digits are the NIC serial number assigned by the manufacturer.

- a. Using the output from the **ipconfig /all** command, answer the following questions.

```
C:\> ipconfig /all
```

```
<output omitted>
```

```
Ethernet adapter Ethernet:
```

```
Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) 82577LM Gigabit Network Connection
Physical Address. . . . . : 5C-26-0A-24-2A-60
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::b875:731b:3c7b:c0b1%10 (Preferred)
IPv4 Address. . . . . : 192.168.1.147 (Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Friday, September 6, 2019 11:08:36 AM
Lease Expires . . . . . : Saturday, September 7, 2019 11:08:36 AM
Default Gateway . . . . . : 192.168.1.1
```

```
<output omitted>
```

What is the OUI portion of the MAC address for this device?

5C-26-0A

What is the serial number portion of the MAC address for this device?

24-2A-60

Using the example above, find the name of the vendor that manufactured this NIC.

Dell Inc.

- b. From the command prompt on PC, issue the **ipconfig /all** command and identify the OUI portion of the MAC address for the NIC of the PC.

00-0C-29

Identify the serial number portion of the MAC address for the NIC of PC.

FF-0B-ED

Identify the name of the vendor that manufactured the NIC of the PC.

VMware

Step 1: Analyze the MAC address for the S1 F0/6 interface.

You can use a variety of commands to display MAC addresses on the switch.

- a. Console into S1 and use the **show interfaces vlan 1** command to find the MAC address information. A sample is shown below. Use output generated by your switch to answer the questions.

```
S1# show interfaces vlan 1
Vlan1 is up, line protocol is up
  Hardware is EtherSVI, address is 001b.0c6d.8f40 (bia 001b.0c6d.8f40)
  Internet address is 192.168.1.2/24
  MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive not supported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input never, output 00:14:51, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicasts)
    0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    34 packets output, 11119 bytes, 0 underruns
    0 output errors, 2 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out
```

What is the MAC address for VLAN 1 on S1?

0024.50ad.07c0

What is the MAC serial number for VLAN 1?

Ad.07c0

What does bia stand for?

Burned In Address

Why does the output show the same MAC address twice?

It shows the logical MAC address and the BIA to ensure no confusion is the logical MAC is spoofed and changed.

- Another way to display the MAC address on the switch is to use the **show arp** command. Use the **show arp** command to display MAC address information. This command maps the Layer 2 address to its corresponding Layer 3 address. A sample is shown below. Use output generated by your switch to answer the questions.

```
S1# show arp
```

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.1.2	-	001b.0c6d.8f40	ARPA	Vlan1
Internet	192.168.1.3	0	5c26.0a24.2a60	ARPA	Vlan1

What Layer 2 addresses are displayed on S1?

Layer 2 shows the MAC addresses

What Layer 3 addresses are displayed on S1?

Layer 3 shows the IP addresses

Step 2: View the MAC addresses on the switch.

Issue the **show mac address-table** command on S1. A sample is shown below. Use output generated by your switch to answer the questions.

```
S1# show mac address-table
```

Mac Address Table

```
-----
```

Vlan	Mac Address	Type	Ports
----	-----	-----	-----
All	0100.0ccc.cccc	STATIC	CPU
All	0100.0ccc.cccd	STATIC	CPU
All	0180.c200.0000	STATIC	CPU
All	0180.c200.0001	STATIC	CPU
All	0180.c200.0002	STATIC	CPU
All	0180.c200.0003	STATIC	CPU
All	0180.c200.0004	STATIC	CPU
All	0180.c200.0005	STATIC	CPU
All	0180.c200.0006	STATIC	CPU
All	0180.c200.0007	STATIC	CPU
All	0180.c200.0008	STATIC	CPU
All	0180.c200.0009	STATIC	CPU
All	0180.c200.000a	STATIC	CPU
All	0180.c200.000b	STATIC	CPU
All	0180.c200.000c	STATIC	CPU
All	0180.c200.000d	STATIC	CPU
All	0180.c200.000e	STATIC	CPU
All	0180.c200.000f	STATIC	CPU
All	0180.c200.0010	STATIC	CPU
All	ffff.ffff.ffff	STATIC	CPU
1	5c26.0a24.2a60	DYNAMIC	Fa0/6

Total Mac Addresses for this criterion: 21

Did the switch display the MAC address of the PC? If you answered yes, what port was it on?

Yes the MAC Address was shown on port Fa0/1

Reflection Questions

Can you have broadcasts at the Layer 2 level? If so, what would the MAC address be?

1. Why would you need to know the MAC address of a device?

You would need to know the MAC address of a device so you can verify that the frames are being transmitted to the correct device. The intermediary devices need to know the MAC addresses so they can send the frames to the correct intermediary and end devices as well.